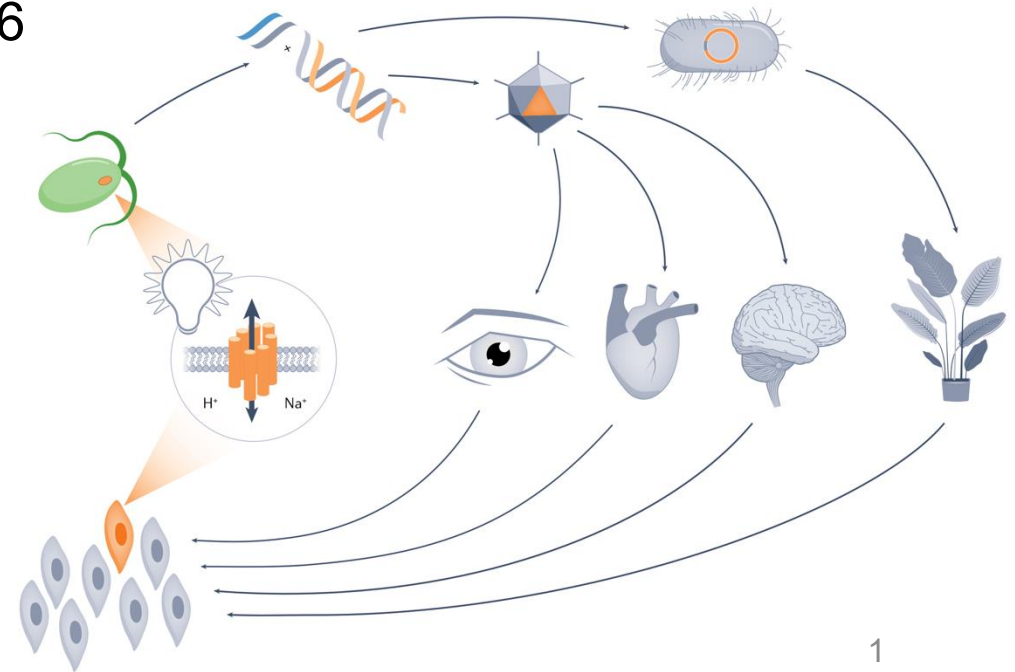


Optogenetic investigation of neural circuits in vivo

Elias Leiva – Universidad de Santiago
FALAN LTPN 2026



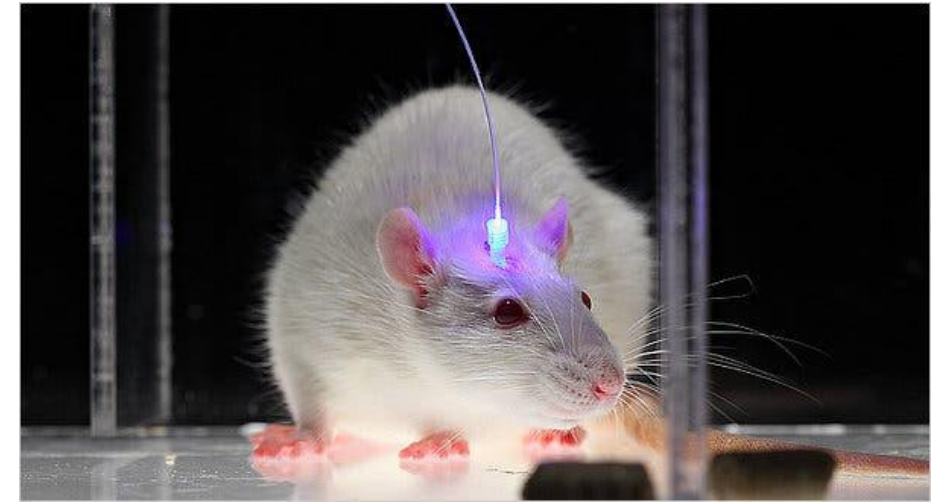
- Principles underlying optogenetic manipulation of neural activity.
- How to select an appropriate opsin, viral strategy, and light-delivery system.
- How to design an in vivo experiment to test circuit connectivity or circuit function.
- Essential controls and major technical limitations.
- How to interpret behavioral and physiological results without overestimating circuit causality.

In vivo optogenetics



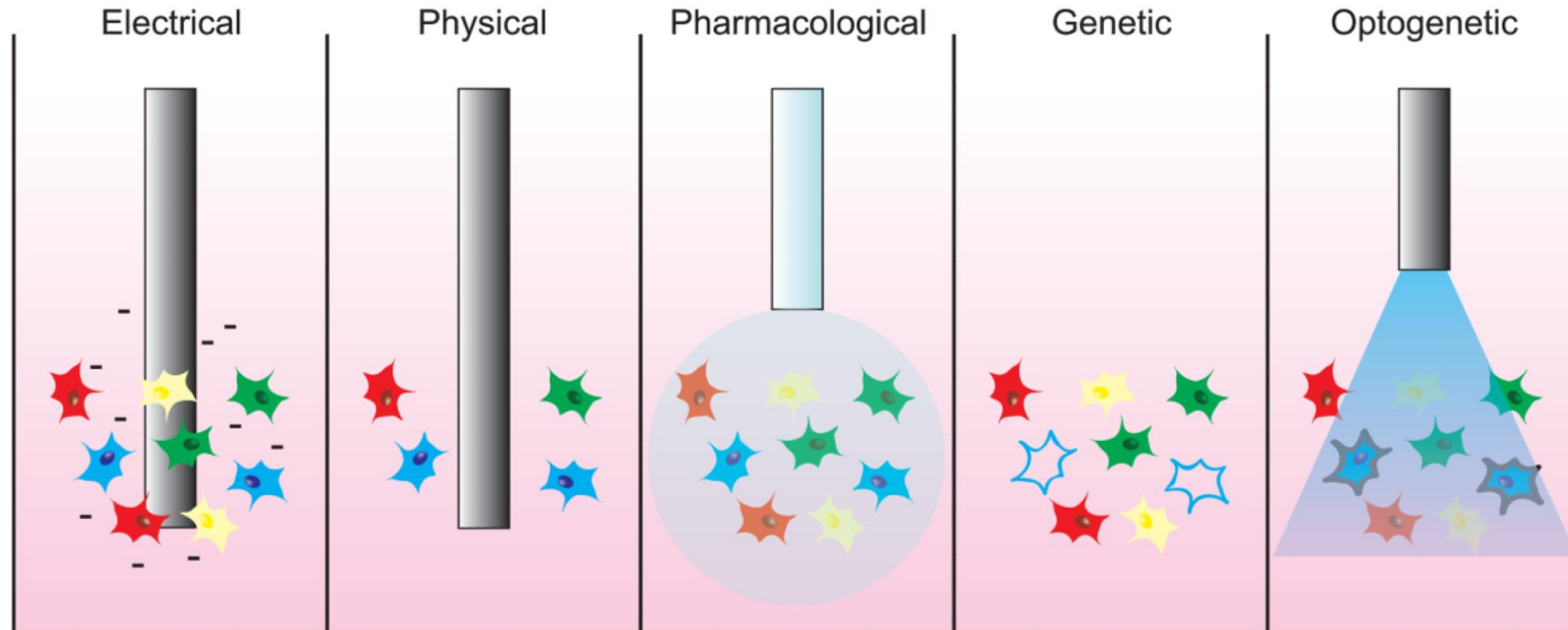
Why optogenetics for in vivo circuit analysis?

- From observing activity to testing causality

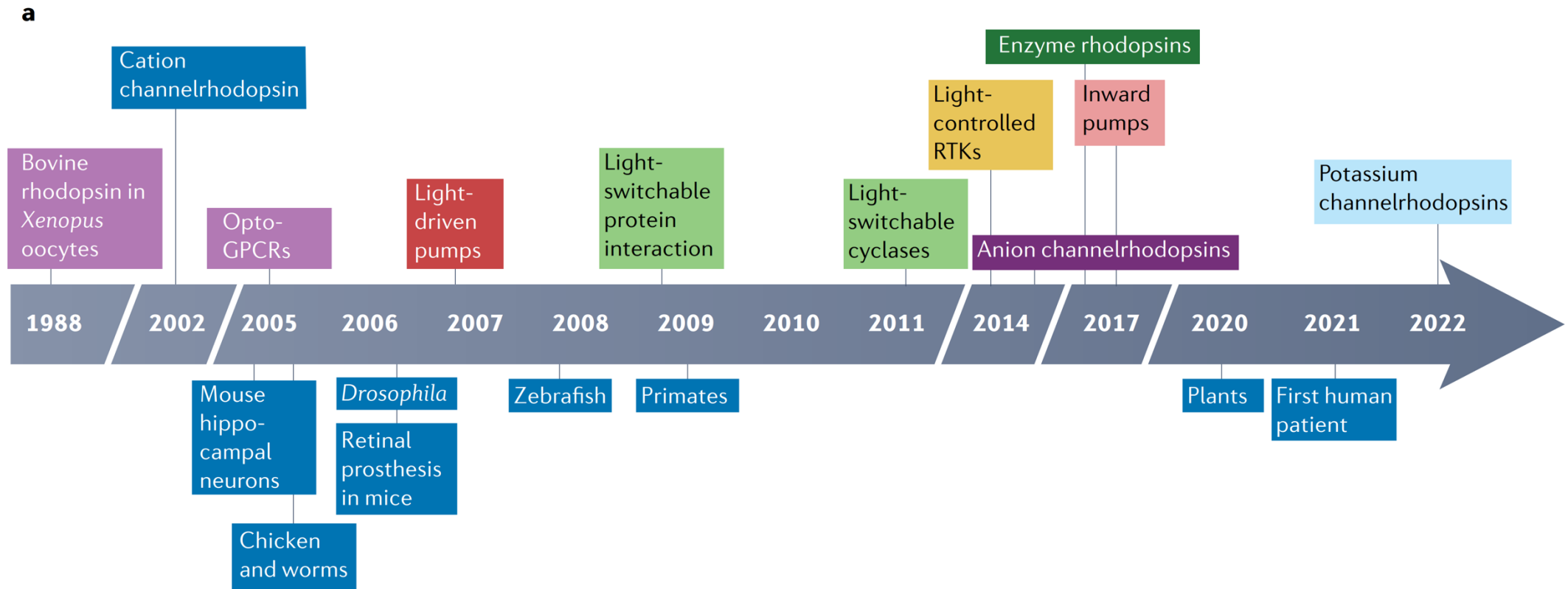


- Sufficiency: activation produces or biases an outcome.
- Necessity: inhibition prevents or weakens an outcome.
- Temporal contribution: the circuit matters during a defined epoch.

Ways to control neuronal behavior



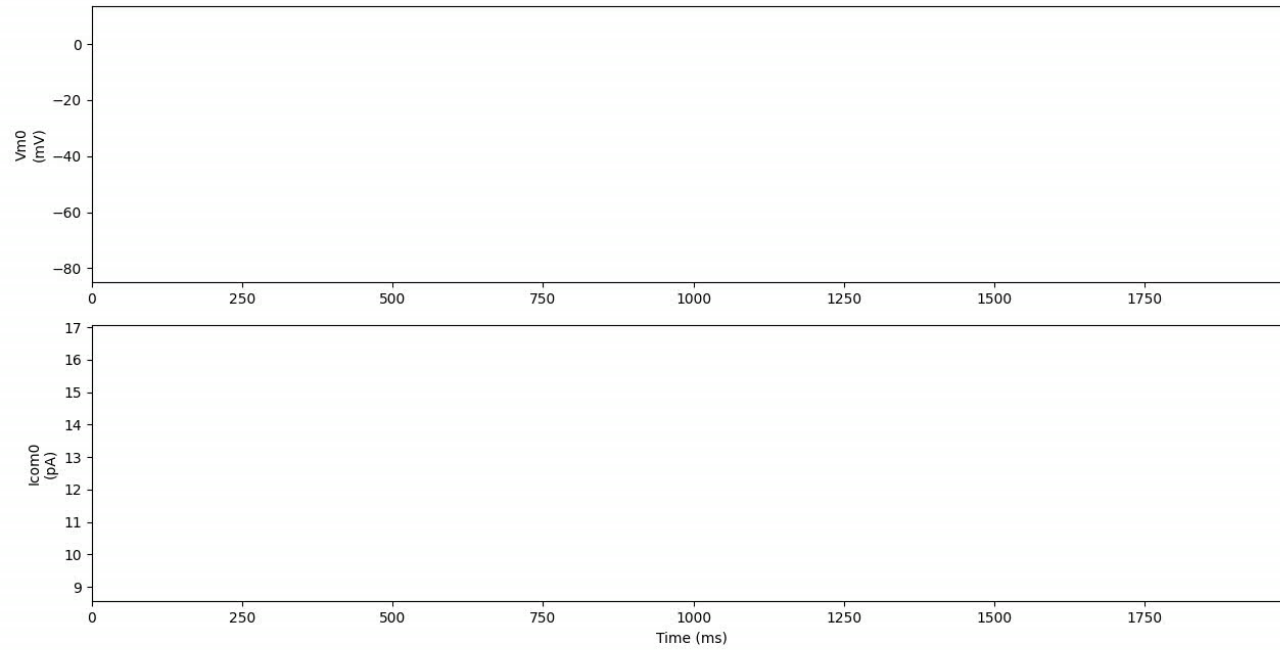
Timeline of the development of optogenetics technique



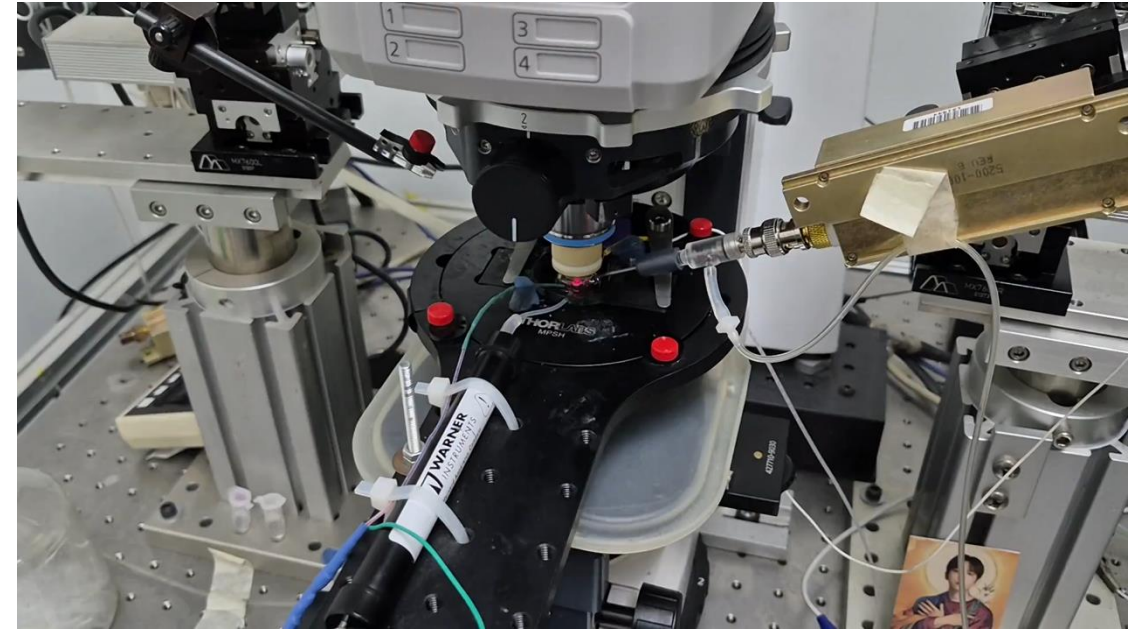
Navarro and Monzo, 2010, DOI:10.3349/ymj.2010.51.5.622

Ex vivo optogenetics

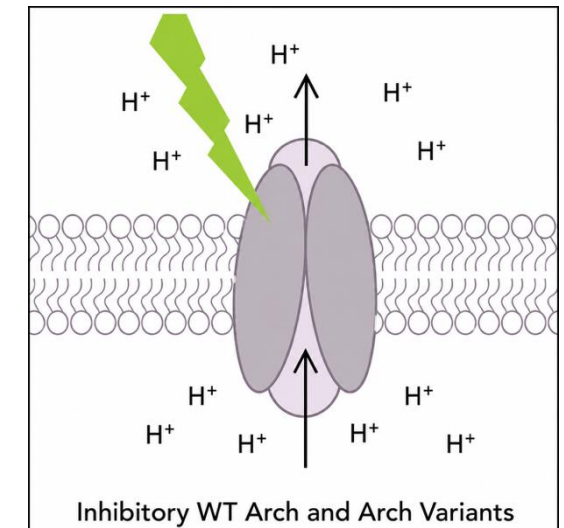
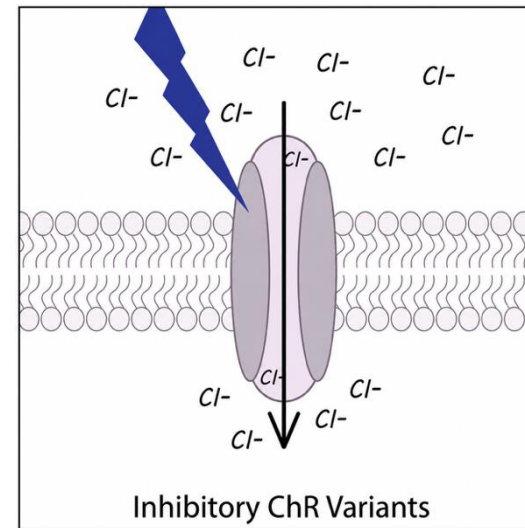
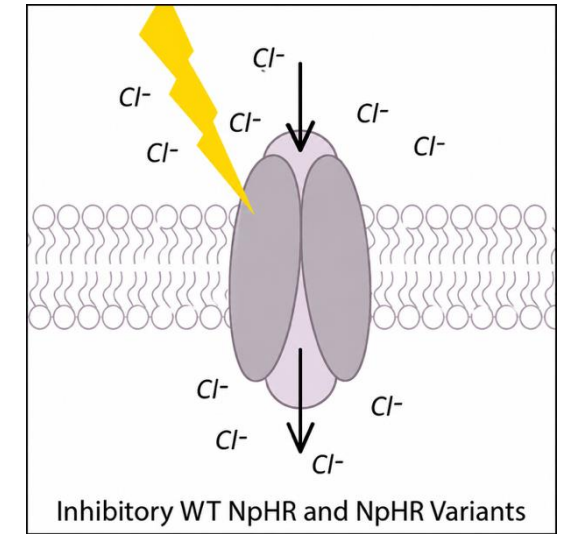
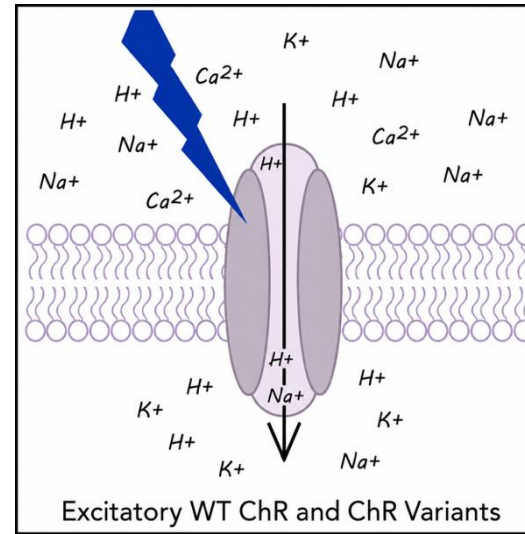
251209_001.Opto_IC_10Hz.1.wcp | Record 1/10



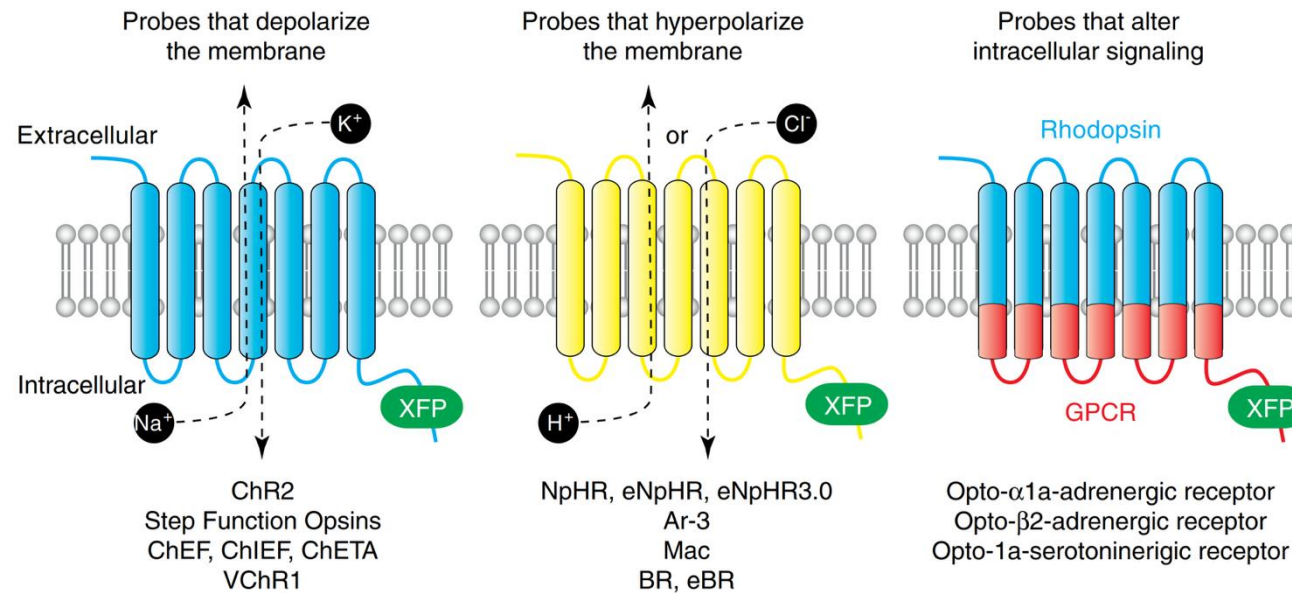
t = 0.00 ms playback = 1x



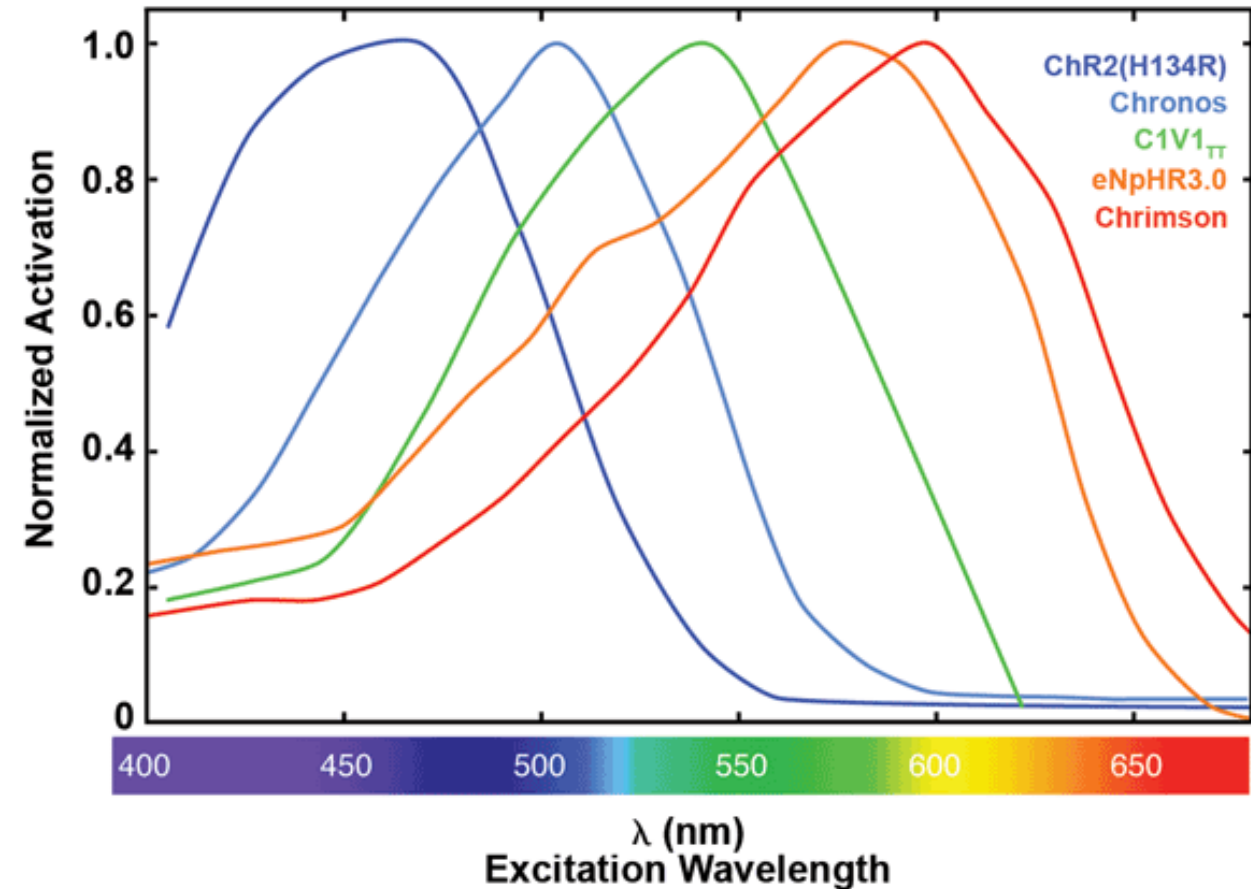
- Opsin Families & Mechanisms:
 - Light activated proteins
 - Respond to specific wavelength
 - Two main group
 - Excitatory
 - Non-selective cation channel
 - Inhibitory
 - Anion channelrhodopsin
 - Chloride pumps
 - Chloride channels
 - Proton pumps
 - Potassium channels



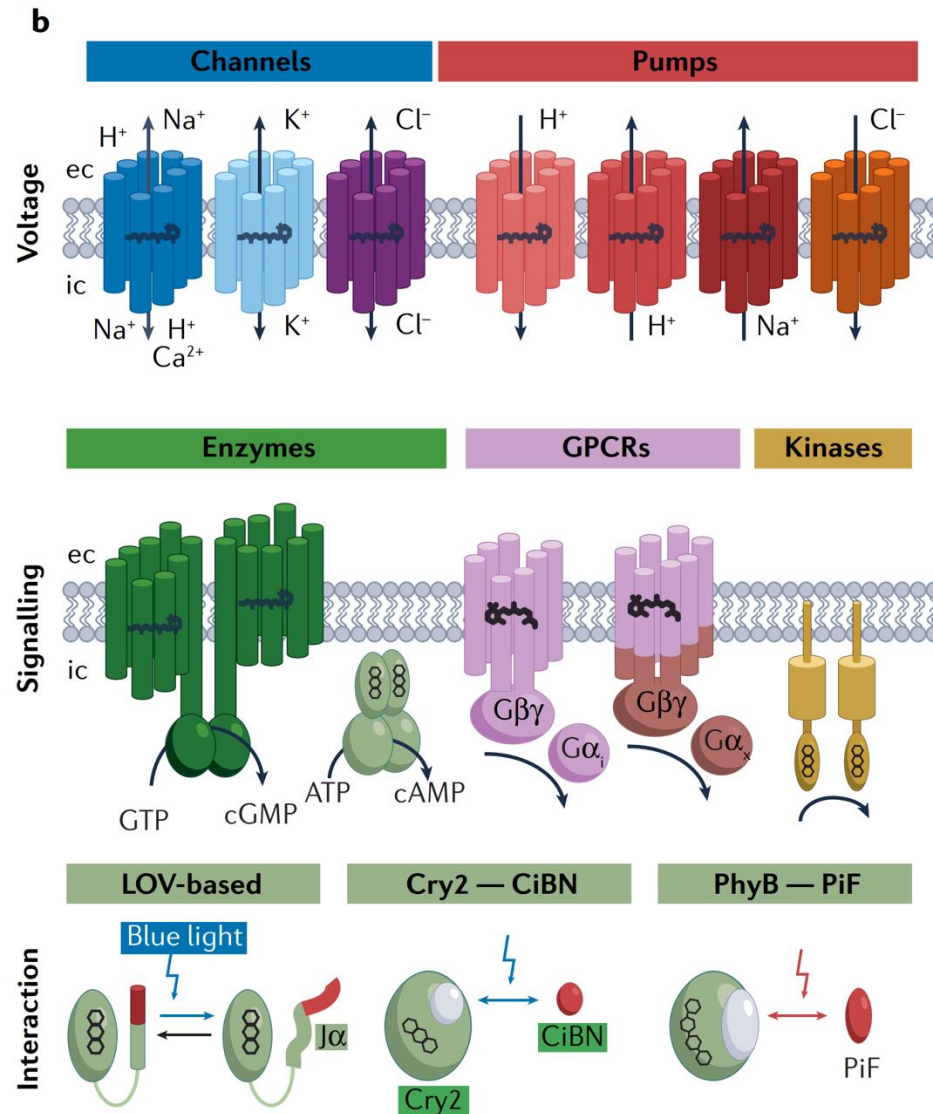
- *Excitatory*: Channelrhodopsin variants (ChR2, ChETA, Chronos for high-frequency driving).
- *Inhibitory*: Halorhodopsin (NpHR; Cl^- pump), Arch/ArchT (H^+ pump), anion channelrhodopsins (GtACR1/2).
- *Biochemical/G-protein*: OptoXRs (Gq-, Gi-, Gs-coupled signaling).



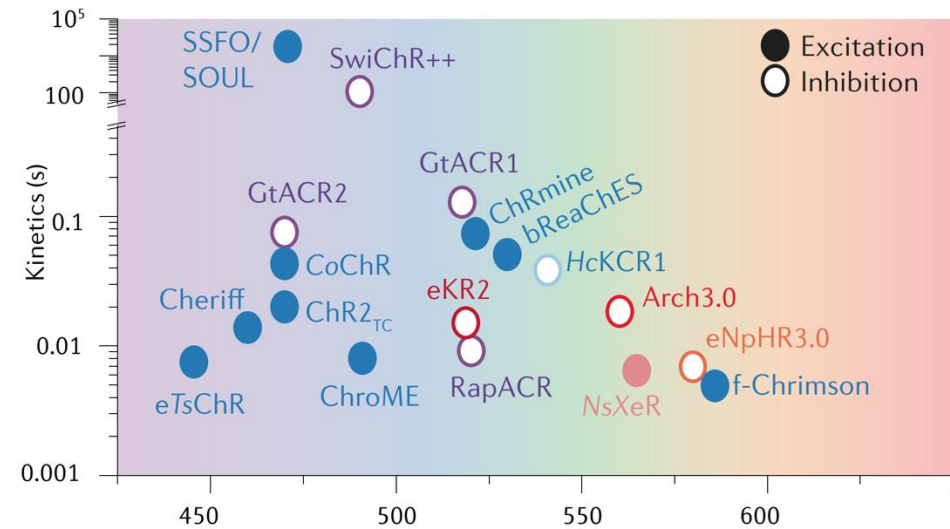
- **Key Photophysical Parameters:**
- Spectral overlap, action spectra, irradiance thresholds (mW/mm^2).
- Kinetics: On/off rates, deactivation constants, and photocurrent desensitization.



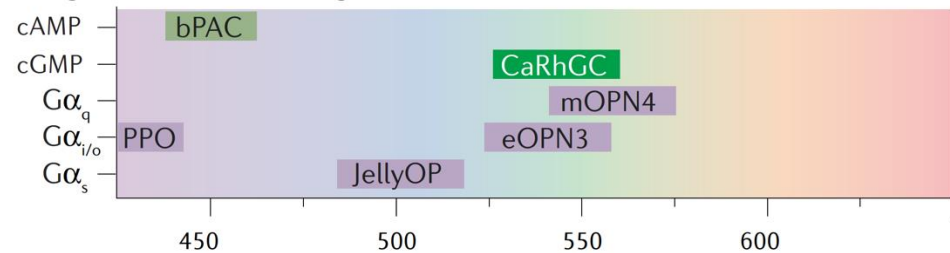
Types of opsins and their functions



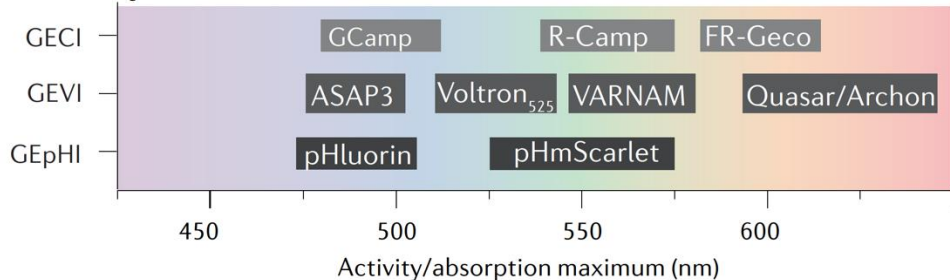
c Microbial opsins



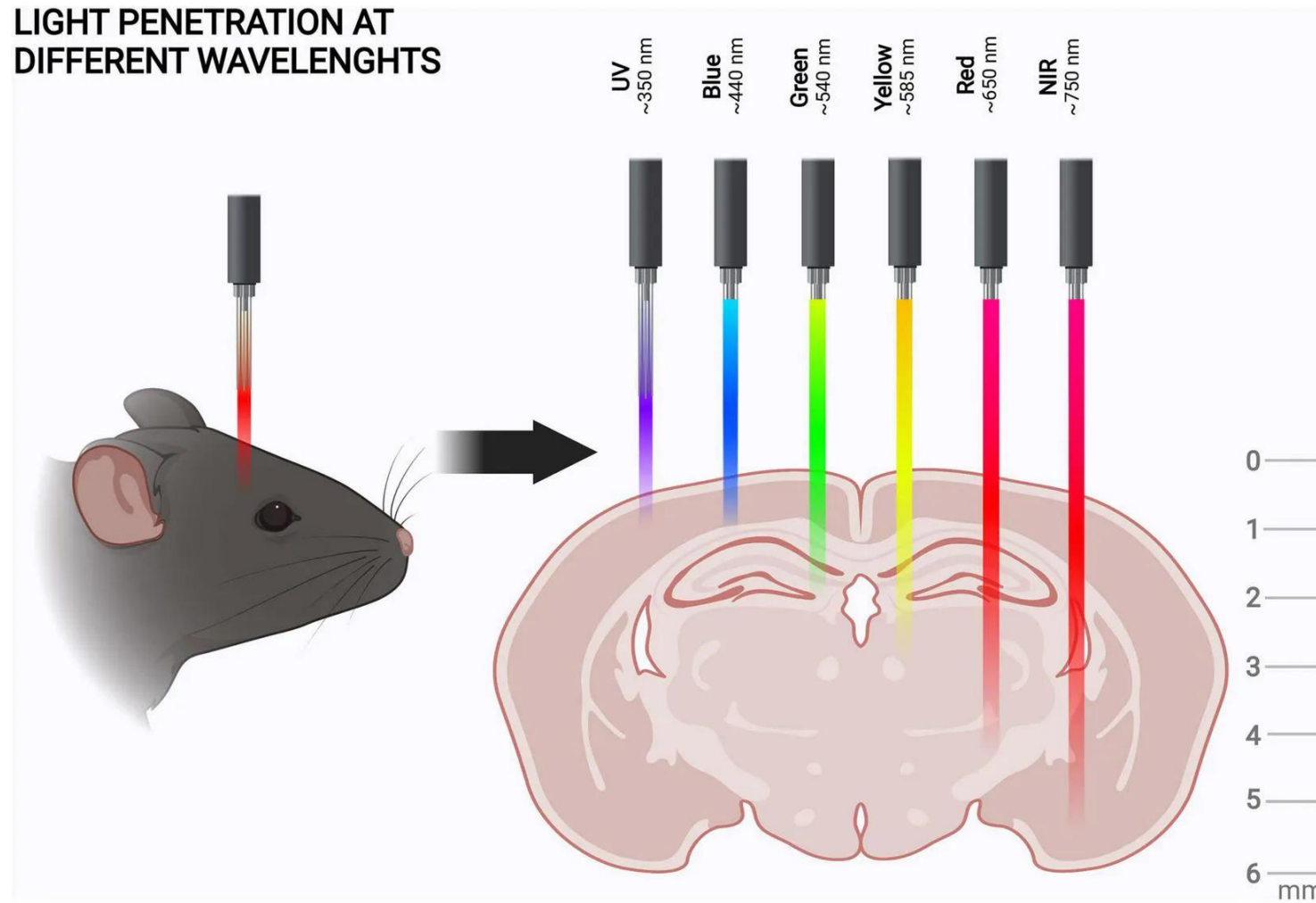
d Signal transduction regulators



e Activity sensor



Light penetration



Main types commercially available opsins

Excitatory opsins

Name	Description	Peak excitation λ (nm)
ChR2	Widely used light-gated cation channel from <i>Chlamydomonas reinhardtii</i> (CrChR2)	470
ChR2/H134R	Widely used variant with larger photocurrent compared to CrChR2	450
ChETA	E123T mutation; faster kinetics but reduced photocurrent amplitude	490
ChR/T159C	T159C mutation; displays increased photocurrents	470
SFO/SSFO	Step Function Opsins and Stabilized Step Function Opsins; mutations at C128 and D156 of CrChR delay the closing of the channel	470 (activation) 590 (inactivation)
ReaChR	Red-activatable variant of CrChR2	590

Main types commercially available opsins

Name	Description	Peak excitation λ (nm)
VChR1	Red-shifted channelrhodopsin from <i>Volvox carteri</i>	570
Chronos	High-speed, red-shifted channel from <i>Stigeoclonium helveticum</i>	500
Chrimson, ChrimsonR	Red-shifted channel from <i>Chlamydomonas noctigama</i> ; ChrimsonR carries K176R mutation with increased on/off kinetics	590
ChRmine	Red-shifted channelrhodopsin from <i>Tiarina fusus</i> ; variants include rsChRmine (further red-shifted), hsChRmine (high speed), and frChRmine (combined high speed and more red-shifted); good response with 585 nm or 650 nm stimulation	~520 (ChRmine, hsChRmine) ~585 (rsChRmine, frChRmine)
PsChR2	High-efficiency blue-shifted channelrhodopsin from <i>Platymonas subcordiformis</i>	445
CoChR	Channelrhodopsin from <i>Chloromonas oogama</i> ; high-efficacy soma-targeted variant soCoChR	470
CheRiff	Blue-shifted channelrhodopsin from <i>Scherffelia dubia</i>	460

Main types commercially available opsins

Inhibitory opsins, K⁺ channel

Name	Description	Peak excitation λ (nm)
C1C2	ChR1-ChR2 chimera	470
ChIEF, ChEF, ChD	Engineered chimeras of ChR1 and ChR2; increased photocurrent amplitude	450–470
C1V1(t/t)	ChR1-VChR1 chimera with E122T and E162T mutations; increased photocurrent amplitude, on/off kinetics, and red-shifted peak action spectra	540
HcKCR	Kalium channelrhodopsins, light-gated K ⁺ channels from <i>Hyphochytrium catenoides</i>	540 (HcKCR1) 490 (HcKCR2)
WiChR	Kalium channelrhodopsin, light-gated K ⁺ channel from <i>Wobblia lunata</i>	490

Main types commercially available opsins

Inhibitory opsins, chloride channel

Name	Description	Peak excitation λ (nm)
GtACR	Chloride-conducting channel from <i>Guillardia theta</i> , used to silence neuronal activity	515 (GtACR1) 470 (GtACR2)
iC++	Artificial anion conducting channelrhodopsin, used to silence neuronal activity	488
iChloC	Chloride-conducting channel, CrChR2 with mutations E90R, D156N, T159C, CE83Q, and E101S; increased photocurrent amplitude	465
SwiChR	Step-waveform inhibitory ChR, chloride channel derived from C1C2; SwiChR _{CA} includes mutation C128A with increased photocurrent amplitude	475
PsChR1	Red-shifted chloride-conducting channel from <i>Proteomonas sulcata</i>	540
Phobos	Blue-shifted iC++ variant; increased photocurrent amplitude	467
Aurora	Red-shifted anion-selective ReaChR variant; increased photocurrent amplitude	517

Main types commercially available opsins

Inhibitory opsins, Chloride pump

Name	Description	Peak excitation λ (nm)
Jaws	Red-shifted, light-driven inward chloride pump from <i>Haloarcula</i> (<i>Halobacterium</i>) <i>salinarum</i> (strain Shark), used to silence neuronal activity	632
Halo/NpHR	Light-driven inward chloride pump from <i>Natronomonas pharaonis</i> , used to silence neuronal activity; Halo is human codon-optimized	589
eNpHR, eNpHR2.0, eNpHR3.0	eNpHR variants with increased photocurrent amplitude; eNpHR3.0 includes trafficking signal from Kir2.1 and ER export signal for improved membrane targeting	589

Main types commercially available opsins

Inhibitory opsins, proton pumps

Name	Description	Peak excitation λ (nm)
Arch	Light-activated outward proton pump from <i>Halorubrum sodomense</i>	566
eArch 3.0	Arch with addition of trafficking signal from Kir2.1; increased photocurrent amplitude	566
ArchT	Light-activated outward proton pump from <i>Halorubrum</i> strain TP009; improved light sensitivity over Arch	566
eArchT 3.0	ArchT with addition of trafficking signal from Kir2.1; increased photocurrent amplitude	566

- **Viral Vectors & Tropism:**

- AAV serotypes (*AAV2/1,2/5,2/9, AAV – retro, AAV – PHP.eB*).
- Retrograde vs. anterograde pathway-specific manipulation.

- **Cell-Type Specificity:**

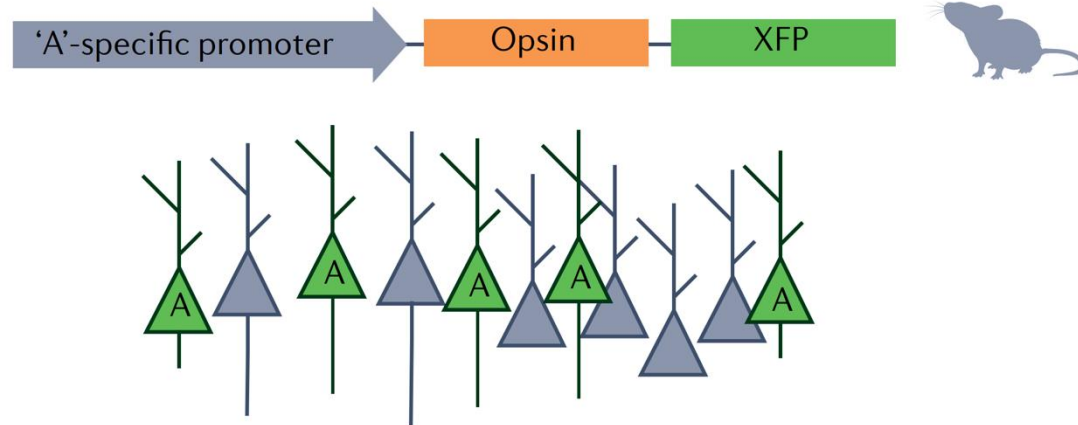
- Cre-LoxP and Flp-FRT recombinase systems (DIO/FLEX constructs).
- Intersectional strategies (Cre + Flp AND/NOT logic gates) to isolate sub-populations.
- Promoter selection: Cell-specific (e.g., CamKIIa, VGAT, PV) vs. ubiquitous (CAG, Synapsin).

Delivery strategies

Strategy	Cell targeting	Time	Key advantages	Main limitations
Transgenic	Cell type-specific promoter	6–12 months	Stable, consistent and relatively uniform expression	Slow; labor-intensive; endogenous promoters may express weakly
Lentivirus	Cell type-specific promoters CaMKII, Synapsin, GFAP, etc	~2 weeks	Rapid testing; strong, long-lasting expression; retrograde variants possible	<10 kb capacity; variable expression; small transduction volume
AAV	Promoter- and serotype-dependent CaMKII, Synapsin, GFAP, etc	~3 weeks	Low immunogenicity; persistent expression; multiple neural serotypes	<5 kb capacity; variable tropism; injection-dependent distribution
Cre-dependent AAV	Determined by Cre-driver line NDNF ^{Cre} , VIP ^{Cre} , SST ^{Cre} , etc	~3 weeks	High specificity and strong expression; DIO minimizes Cre-independent leak	Requires an appropriate Cre line; retains AAV size and injection limits

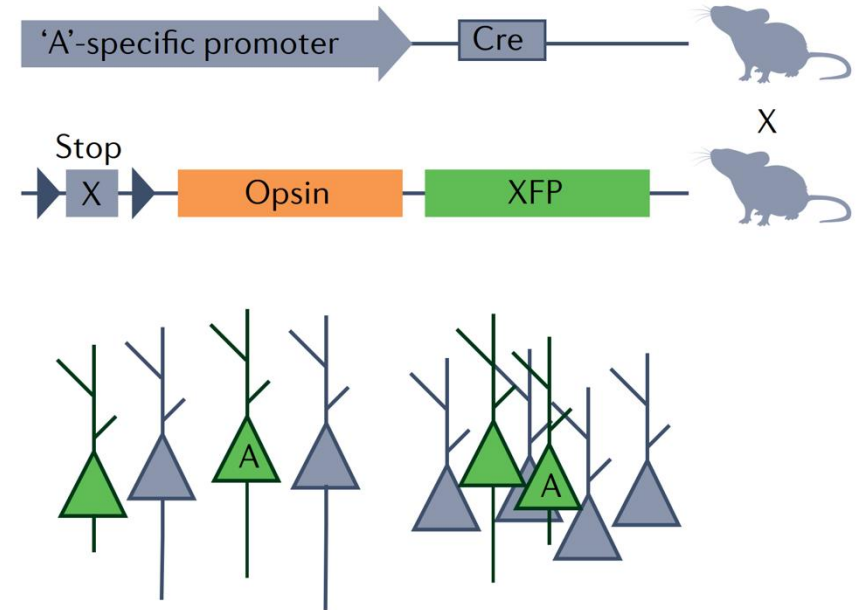
- Animal based
 - Full transgenic
 - Cre-LoxP based

a



Expression: brain-wide, all neurons expressing promoter 'A'

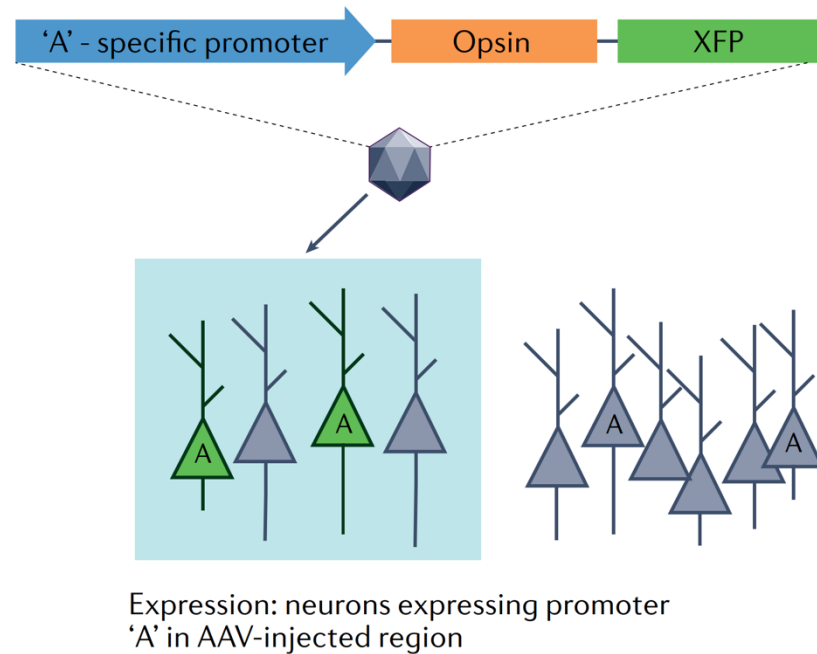
b



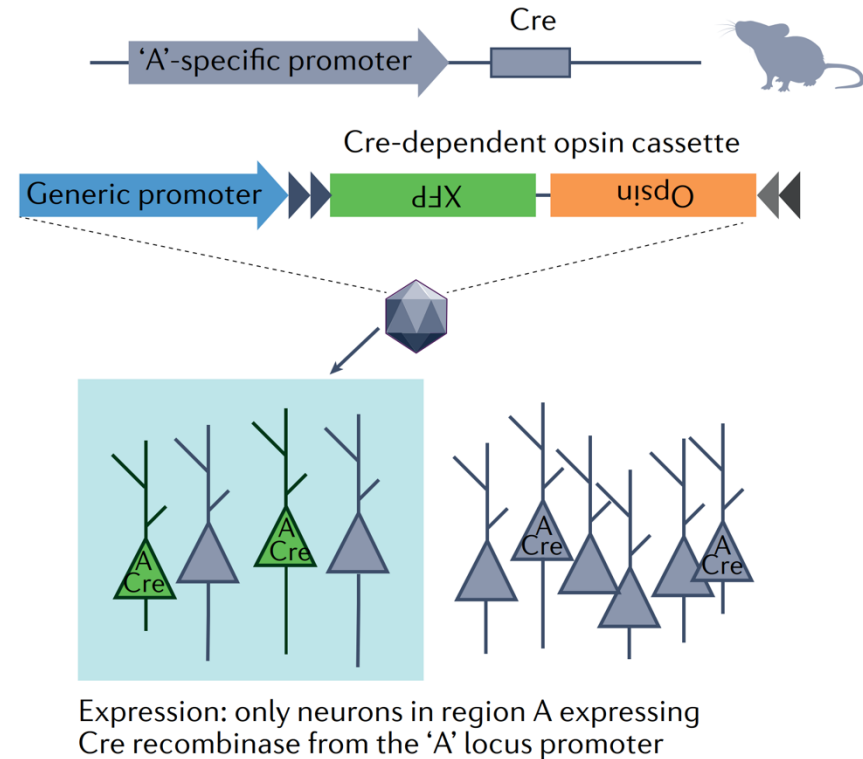
Expression: brain-wide, all neurons expressing promoter 'A'

- Viral injection
 - Promotor-specific based strategies
 - Cre-dependent viruses

c

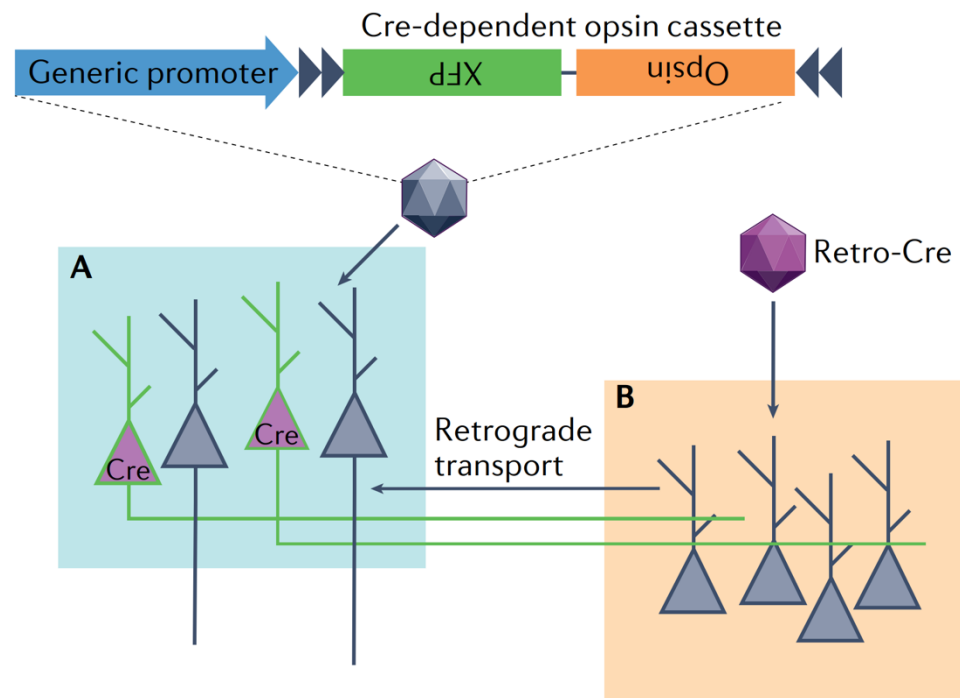


d



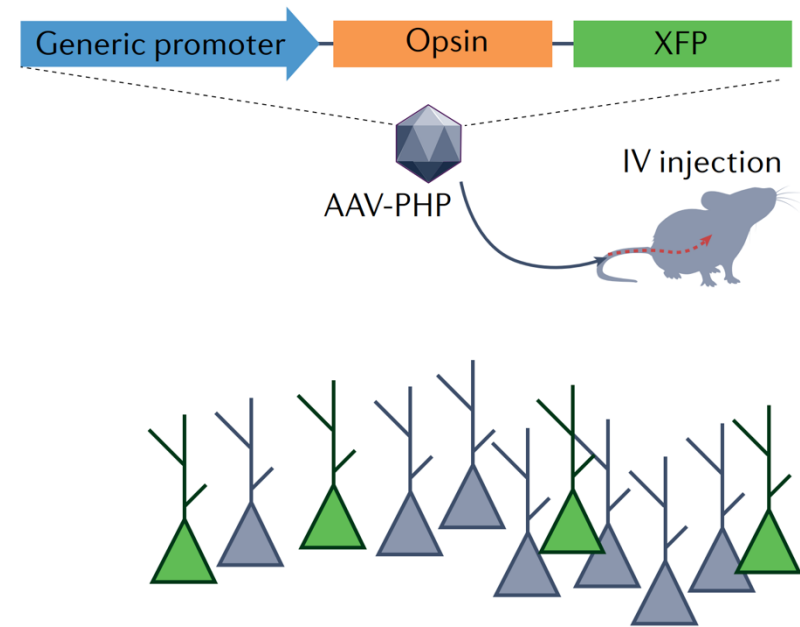
- Retrotracing and bodily injected viruses

e



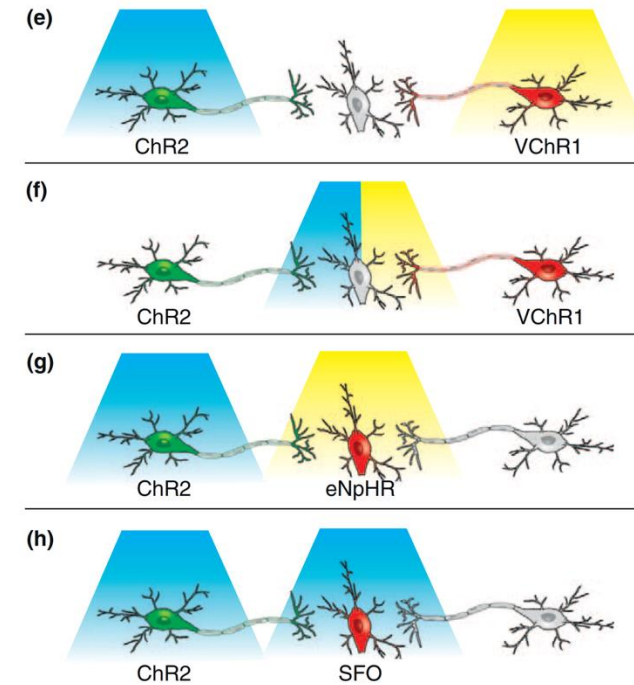
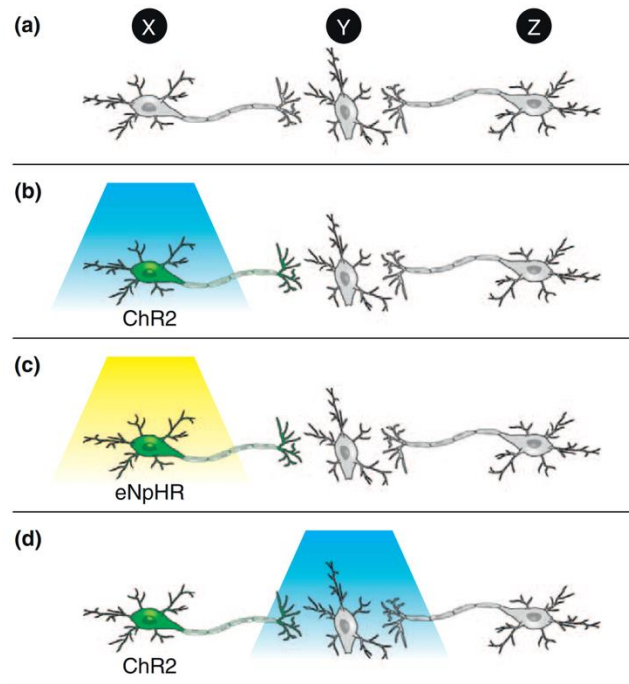
Expression: only neurons in region A projecting to region B

f



Expression: randomly selected brain-wide population
(can be combined with recombinase-conditional vectors)

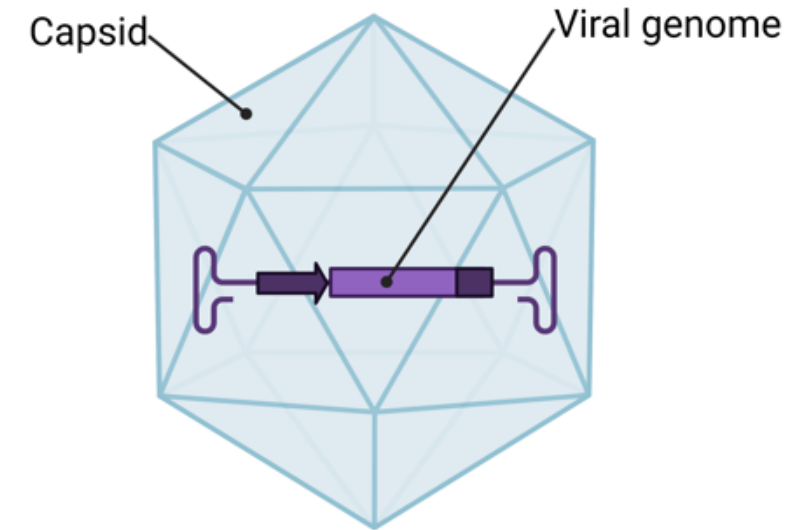
Specific stimulation or inhibition at the same time: Multiplexing



TRENDS in Molecular Medicine

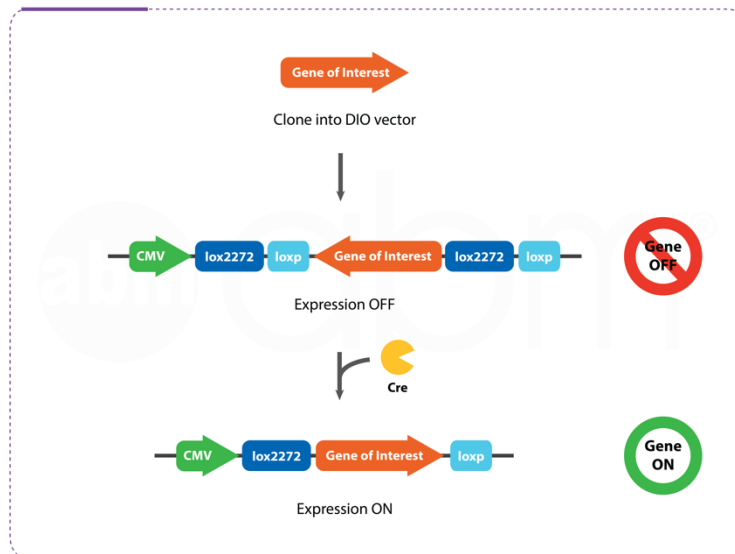
Serotypes, spreading and specificity

Capsid	Primary receptor	Coreceptor
AAV1	• $\alpha 2,3$-/$\alpha 2,6$ N-linked sialic acids¹ • AAV receptor (AAVR)²	
AAV2	• Heparan sulfate proteoglycan (HSPG)³ • AAV receptor (AAVR; not strictly required)²	• Human fibroblast growth factor receptor 1 (FGFR1)³ • $\alpha V\beta 5$ and $\alpha 5\beta 1$ integrins³ • Hepatocyte growth factor receptor (HGFR)³ • Laminin receptor (LR)³ • CD9³
AAV5	• 2,3-N-linked sialic acid⁴ • AAV receptor (AAVR)²	• Platelet-derived growth factor receptor (PDGFR) α and β³
AAV6	• $\alpha 2,3$-/$\alpha 2,6$ N-linked sialic acids¹ • Heparan sulfate proteoglycan (HSPG)³ • AAV receptor (AAVR)²	• Epidermal growth factor receptor (EGFR)³
AAV8	• Laminin receptor (LR)³ • AAV receptor (AAVR)² • Carboxypeptidase D (CPD), also called AAVR2⁵	
AAV9	• Terminal N-linked galactose³ • AAV receptor (AAVR)²	• Laminin receptor (LR)³ • Putative integrin³
AAV11	• Carboxypeptidase D (CPD), also called AAVR2⁵	

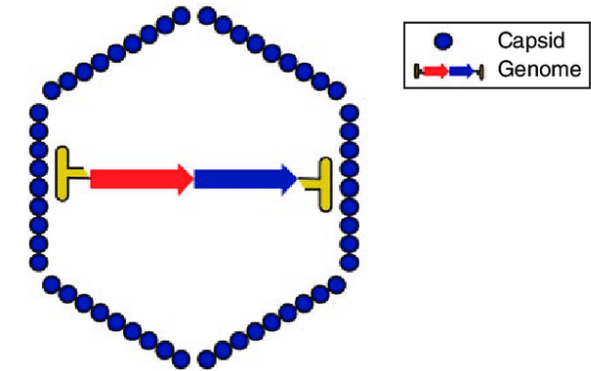


Which AAV serotype to use

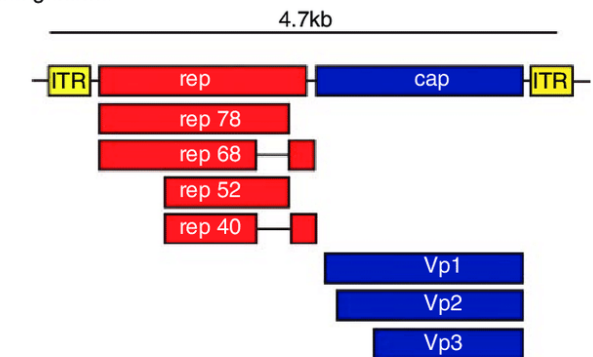
To target...	Try using...
CNS	AAV1, AAV2, AAV2-retro (AAVrg), AAV5, AAV8, AAV9, AAV-PHP.eB, AAV11
Heart	AAV1, AAV8, AAV9
Kidney	AAV8
Liver	AAV8, AAV9
Lung	AAV6
Pancreas	AAV2, AAV8
Photoreceptor cells	AAV2, AAV5, AAV8
Skeletal muscle	AAV1, AAV8, AAV9



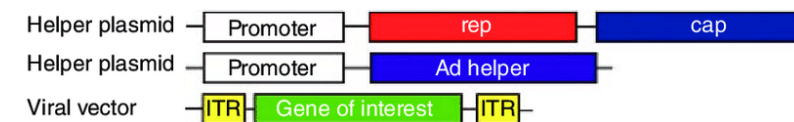
(A) AAV virus scheme



(B) Viral genome



(C) Viral vector system



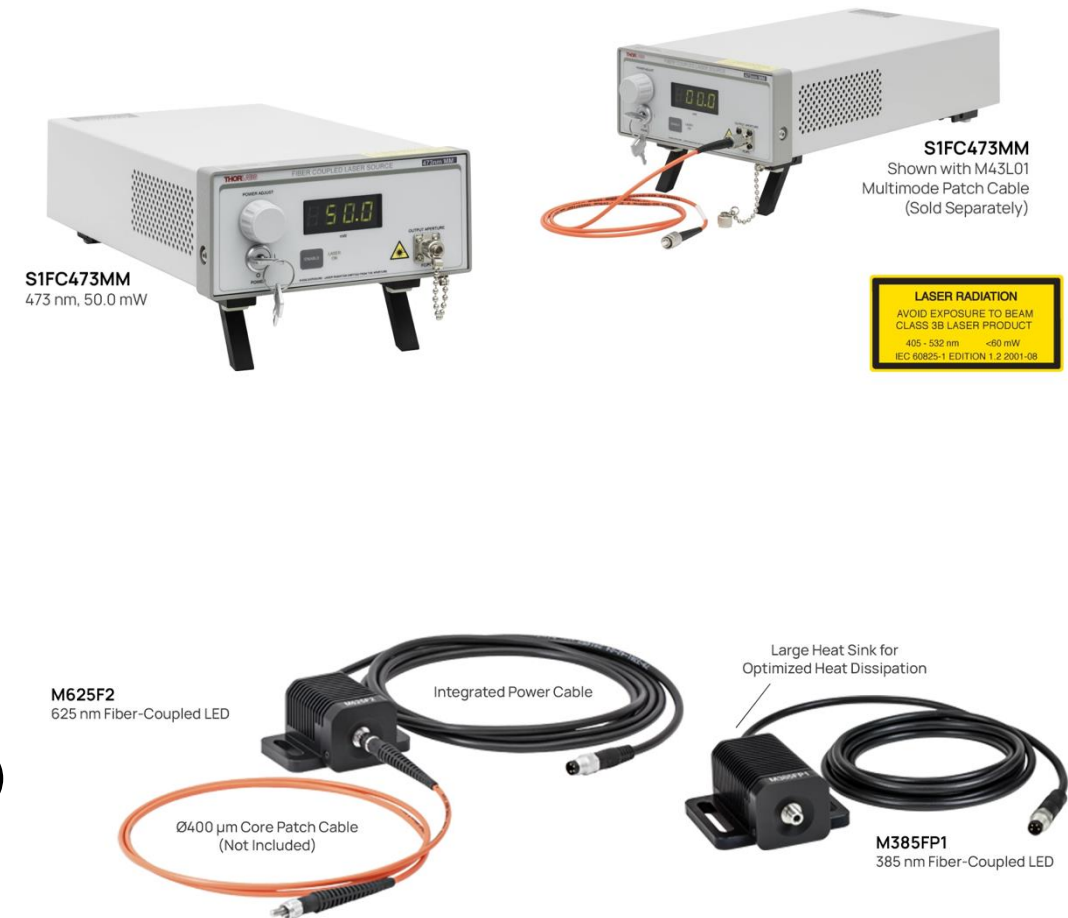
- Light Delivery Systems:

- Laser

- Specific
 - Highly collimated
 - Delivers high power (50 mW or higher)
 - Deep penetration
 - Expensive
 - Tissue thermal problems

- LED

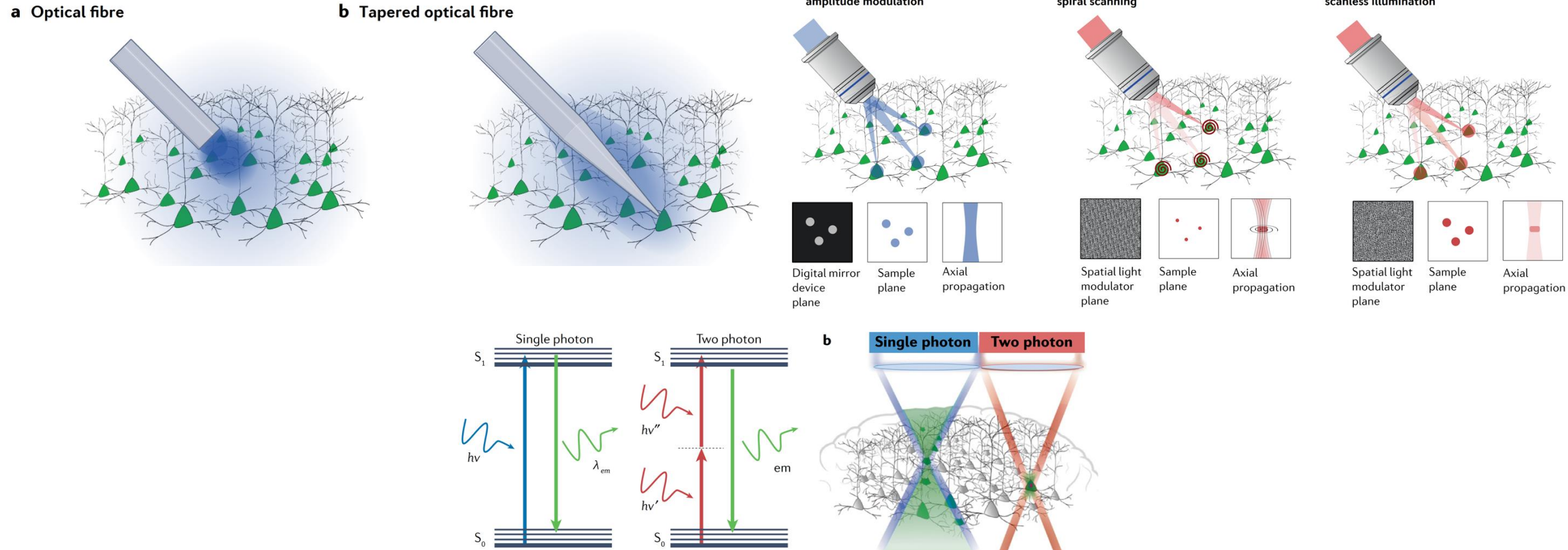
- Specific
 - Scattered light
 - Not able to deliver high power (few mW)
 - Medium penetration
 - Cheap
 - Less prompt to thermal problems



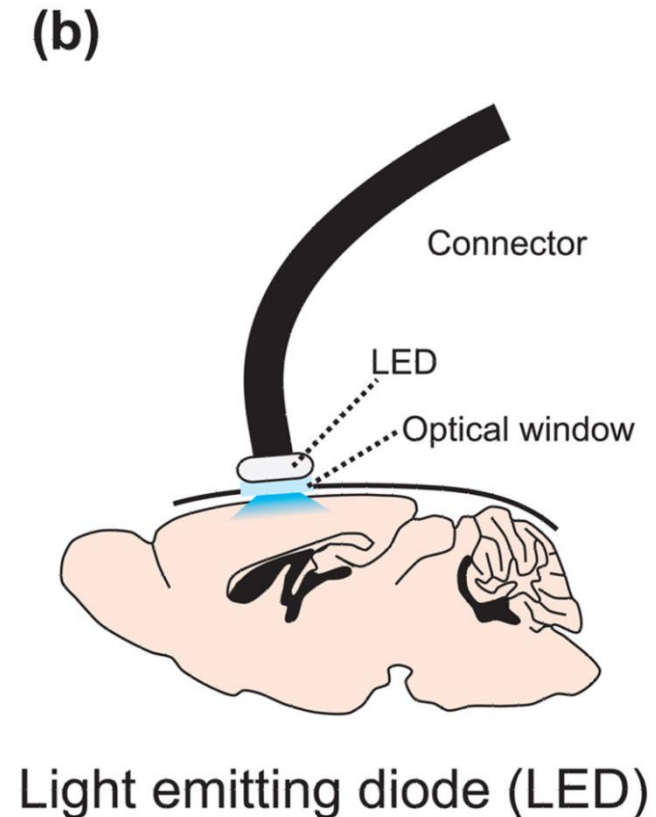
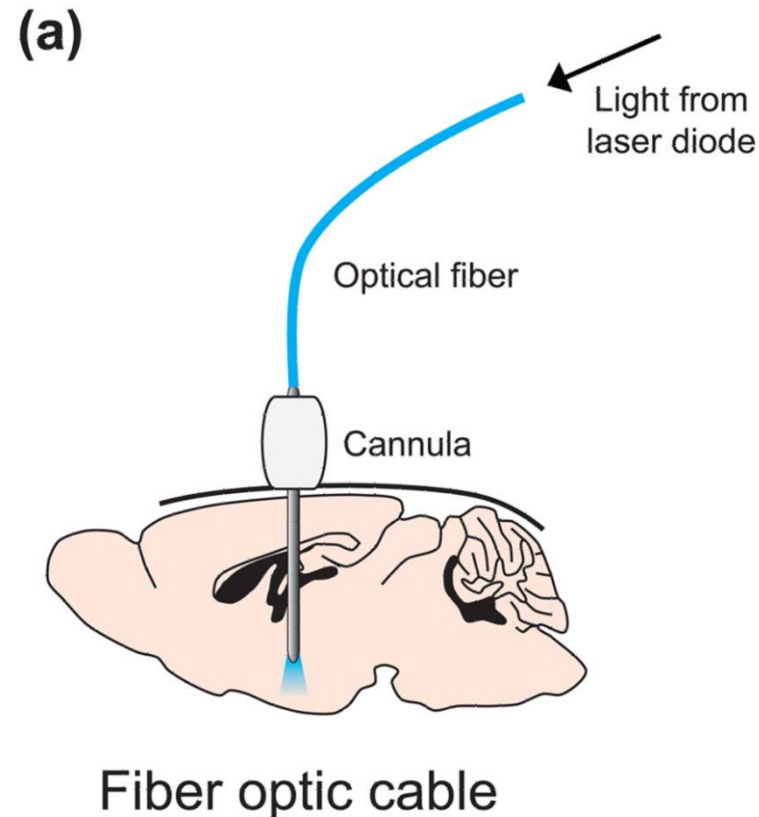
Laser or LED?

Feature	Laser	LED
Light Coherence	Coherent (directional, single phase)	Incoherent (divergent, unphased)
Fiber Coupling Efficiency	High (>50–80%) into micro-fibers	Low (<5–15%) into narrow fibers
Spectral Bandwidth	Extremely narrow (<1 nm)	Moderate (~20–30 nm)
Primary Use Cases	Deep-brain fiber implants, 2-photon/single-cell targeting	Surface/cortical widefield, wireless head-mounted μ LEDs
System Cost	Moderate to Very High (\$5,000 – \$50,000+)	Low to Moderate (\$200 – \$3,000)
Setup Complexity	High (optical alignment, mirrors, safety)	Low (plug-and-play electronic control)

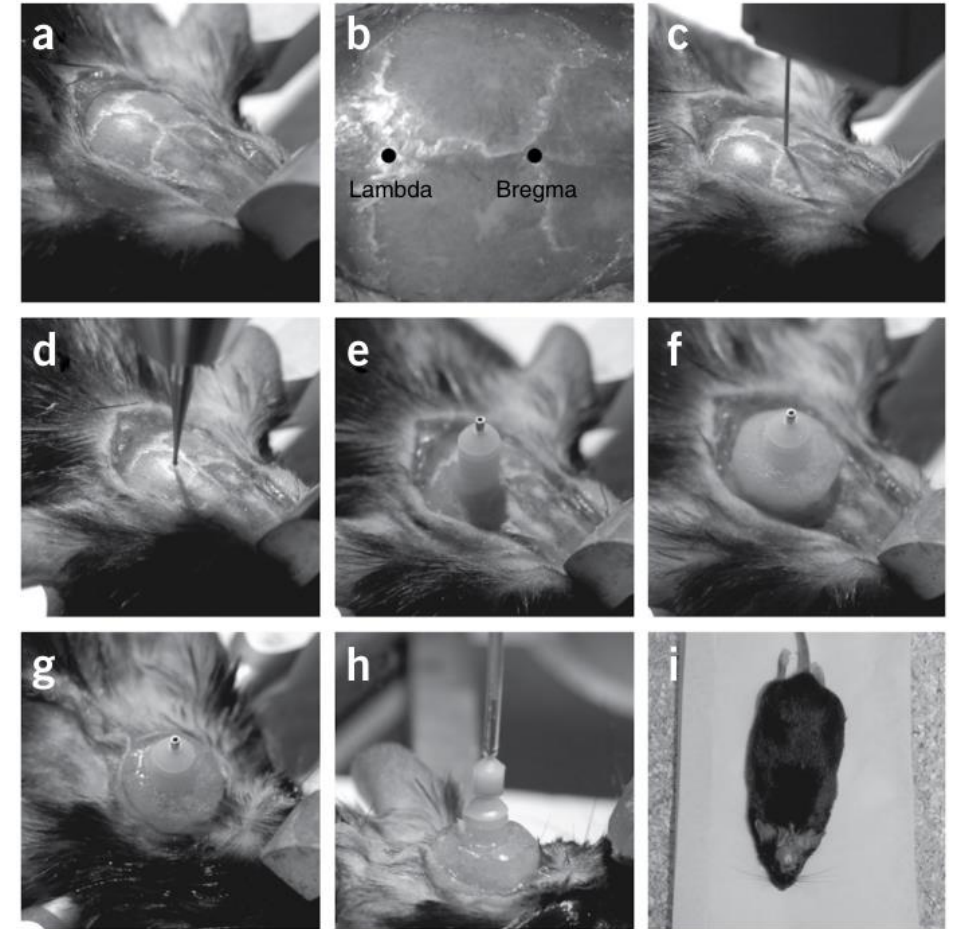
- Light Delivery Systems:



- Types of probe for stimulation
 - Fiber coupled laser
 - Fiber-coupled led
 - Two-photon stimulation

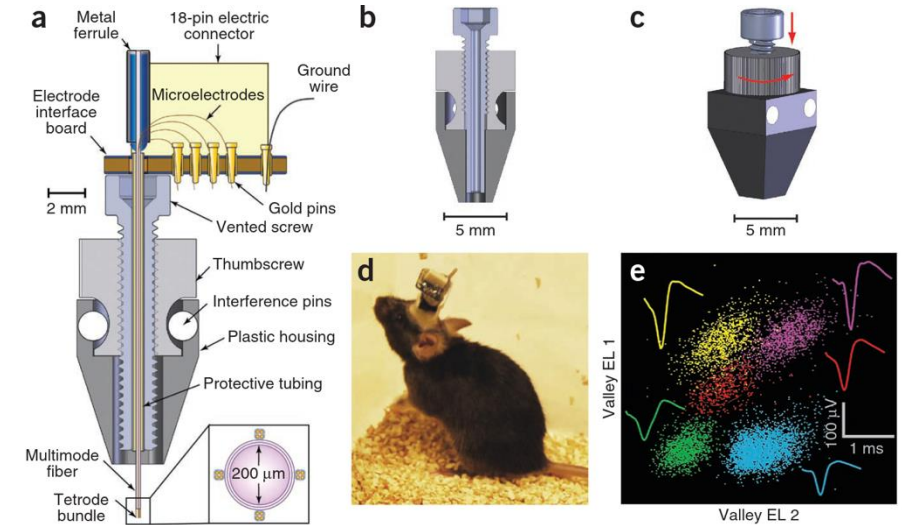


- Light Delivery Systems:
 - Implantable optical fibers
 - Chronic cannulas
 - micro-LEDs (μ LEDs).
 - Single-photon
 - Two-photon optogenetics
 - holographic/Spatial Light Modulators [SLMs] for single-cell resolution.

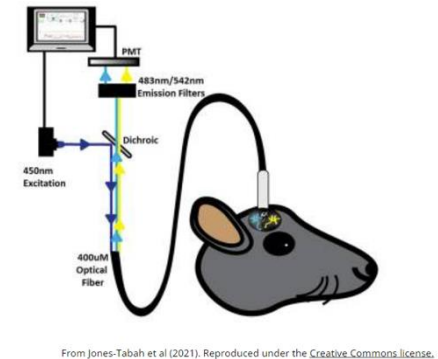
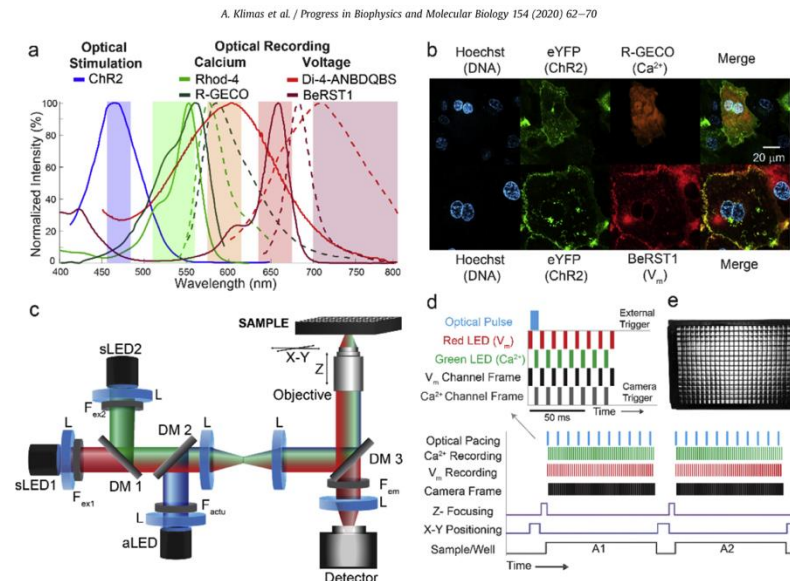


In Vivo Hardware Integration & Readouts

- Simultaneous Readouts:
- Optrode recordings (extracellular spike sorting during light pulses).
- Fiber photometry (combining calcium indicators like jRCaMP1a with red-shifted opsins like Chrimson).
- All-optical electrophysiology and two-photon calcium imaging



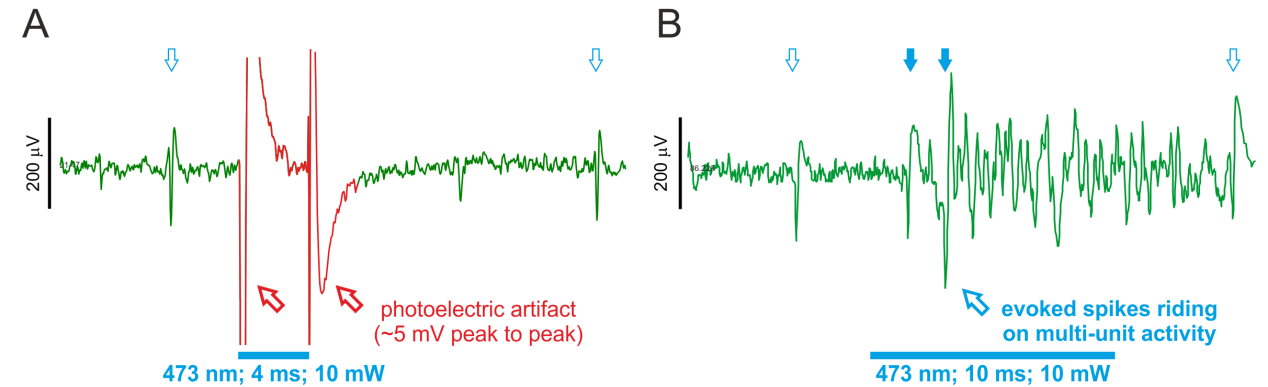
Anikeeva et al 2012, DOI:10.1038/nn.2992



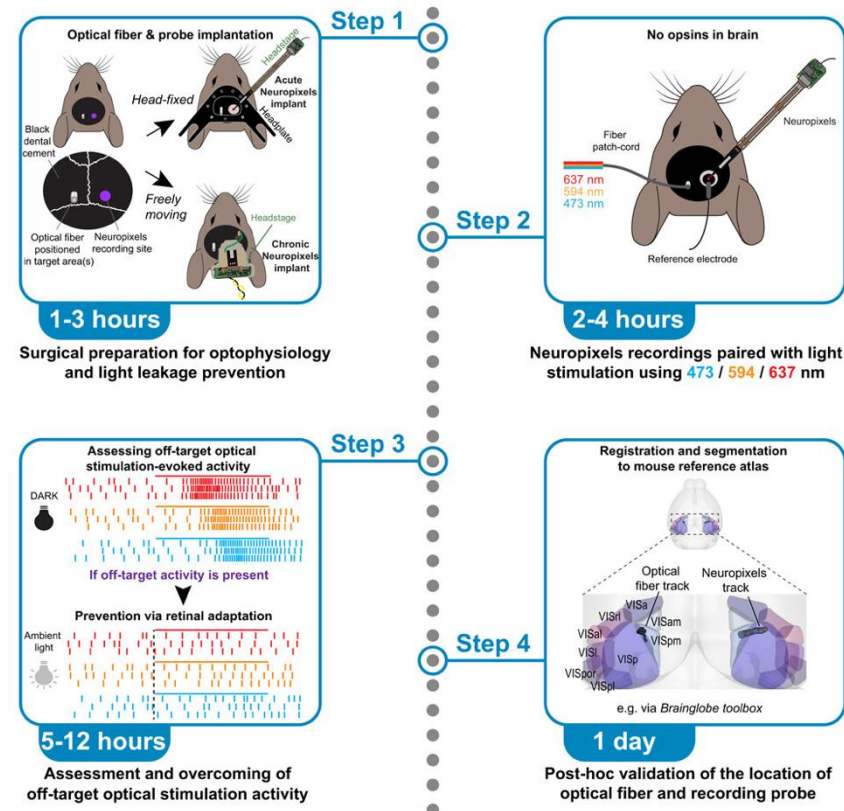
From Jones-Tabah et al (2021). Reproduced under the Creative Commons license.

Klimas et al, 2019, DOI:10.1016/j.pbiomolbio.2019.02.004

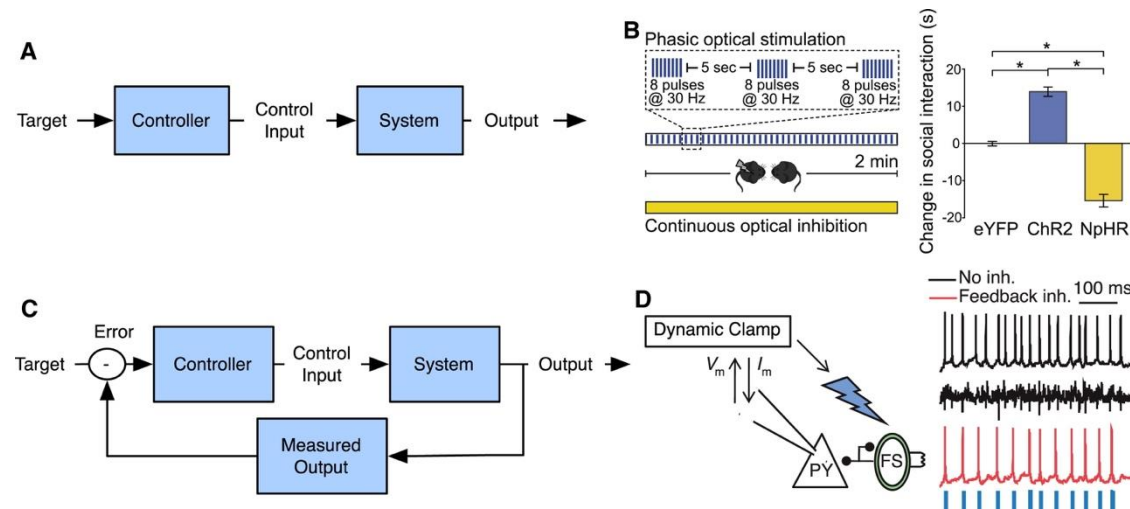
- Technical Artifacts & Physiological disruption
 - Light-induced tissue heating and local vascular responses.
 - Photo-electrophysiological stimulation artifacts on recording electrodes.
 - Altered intracellular chloride reversal potential (E_{Cl}) from prolonged NpHR activation.
 - Rebound excitation following inhibitory light offset.
 - Non-physiological synchronization of population firing.



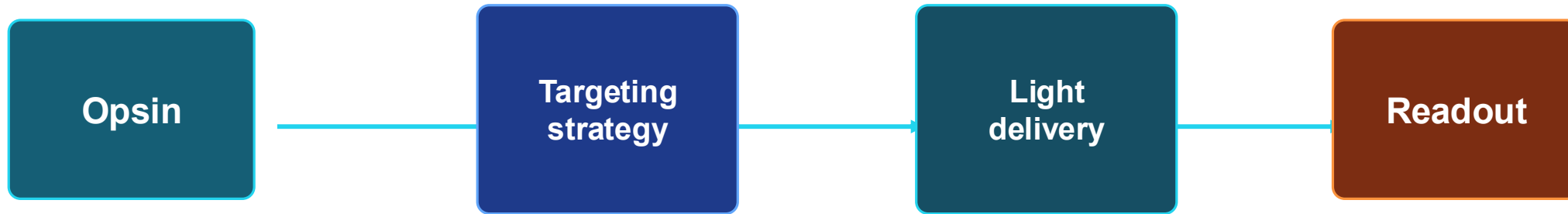
- **Rigorous Experimental Controls:**
 - Fluorophore-only controls (e.g., AAV-DIO-eYFP) at identical irradiance.
 - Off-target anatomical illumination and laser power titrations.



- Closed-Loop Optogenetics:
 - Real-time feedback systems (e.g., interrupting seizure activity upon detection).
- Behavioral Causality:
 - Translating gain-of-function and loss-of-function manipulations into gain/loss of specific behavioral phenotypes.



Architecture of an in vivo optogenetic experiment



Question-driven choices

- Excite, inhibit, or tag?
- Cell body or projection?
- Behavior, spikes, LFP, photometry, imaging?

Interpretability depends on the weakest link

- Weak targeting → weak specificity.
- Weak validation → weak causal claim.
- Wrong epoch → false-negative result.

Typical experimental strategy for circuit dissection

What I must choose before the experiments



Targeting options

- Cre-driver lines
- Cell-type-selective promoters
- Cre/Flp intersectional designs
- Activity-dependent labeling

Core validation

- Injection site
- Fiber position
- Expression specificity
- Functional light response

In vivo optogenetics



- **Software**

- Calman (<https://github.com/flatironinstitute/caiman>)
- Deeplabcut (<https://deeplabcut.github.io/DeepLabCut>)
- efel (<https://ebrains.eu/data-tools-services/tools/efel>)
- Micromanager (<https://micro-manager.org/>)
- WinWCP (https://spider.science.strath.ac.uk/sipbs/software_ses.htm)
- Janelia open science initiative (<https://ossi.janelia.org/projects/>)
- Janelia

- **Animals, viruses, hardware**

- Jackson laboratories (<https://www.jax.org/research-and-faculty/resources/optogenetics-resource>)
- Addgene (<https://www.addgene.org>)
- Open behavior (<https://edspace.american.edu/openbehavior/>)

- **Technical forums and blogs**

- Neuromethods (<https://neuromethods.zulipchat.com/login/>)
- Labrigger (<https://labrigger.com/blog/>)
- GCamp6 (<https://gcamp6f.com/>)
- A neuroscientific blog (<https://neurobiology.substack.com/>)
- Diesseroth lab (<https://dlab.stanford.edu/resources/optogenetics/hardware-setup>)
- Boyden lab (<https://synthneuro.org/>)